MLFF Architecture Revision 48

SIZE-HALVE1 correction, SIZE-FIDELITY1 calibration, and PERF-P2R roadmap

mdstats project

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Revision 48 decision

Revision 48 corrects the generated target-size selection rule discovered during PERF-P2 qualification. Coverage is now **hard admission only**. It no longer selects the four smallest passing sizes.

The corrected authority is

$$7 \xrightarrow{\text{hard coverage}} N_{\text{eligible}} \xrightarrow{\text{3 epochs}} \leq 4 \xrightarrow{\text{10 epochs}} 2 \xrightarrow{\text{30 epochs}} 1.$$

The low-fidelity resource-allocation pattern is related to successive halving [1,2], while mdstats retains its own deterministic target metrics, hard gates, provenance, and exact restart semantics.

Why the correction is required

The former Stage-A rule treated passing geometric/statistical coverage as sufficient evidence that larger nested data sets could not improve training. That implication is not valid. A small subset may span the required support yet sample important regions too sparsely for an accurate learned potential.

Coverage therefore answers only whether a candidate is admissible. Training response answers whether its sampling density/information content is adequate.

Authority changes

Revision 48 advances current generated authority to:

- TARGET-DATA2C v3: every globally materializable rung is materialized;
- TARGET-DATA2D v2: hard coverage plus exact 3/10/30 successive fidelity;
- target-size training evidence v3: checkpoint, optimizer/scheduler, and RNG continuation ancestry;
- TARGET-DATA2E v2: complete coarse/short/final funnel provenance; and
- EVAL2 `size_development_coarse`: fixed common target-only epoch-3 role.

Historical TARGET-DATA2C v1/v2 and PERF-P2 evidence remain archival. They are stale as current generated-campaign authority.

Additional safeguards found during re-audit

Coarse screen must be past warm-up

Generated preflight now requires

$$E_3/E_{30} > p_{\text{warmup,end}}.$$

The defaults satisfy $3/30 = 0.10 > 0.05$. A screen inside warm-up is rejected.

Common coarse evaluation role

The epoch-3 target monitor is a versioned policy field, default 256 configurations. It is a deterministic, correlation-block-balanced subset of the same leakage-safe development complement used by the full size study. Every candidate is evaluated on identical target frames. Replay inference is omitted at epoch 3.

Exact continuation includes RNG state

Promotion is an exact continuation, not retraining. The authority authenticates checkpoint/model, optimizer/scheduler, and Python/NumPy/Torch CPU/CUDA RNG state at both 3->10 and 10->30 boundaries. Evidence also records optimizer updates and structures presented because equal epoch counts do not imply equal compute exposure across data sizes.

Early equivalence preserves the ladder boundary

Smaller-size preference is retained only at final 30-epoch qualification. During epoch-3 and epoch-10 halving, the largest hard-coverage boundary is preserved **within its practical-equivalence band**. It may still be eliminated when materially worse. This keeps the workflow capable of detecting `nonconverged_at_ladder_boundary` rather than deleting the boundary on an early tie.

SIZE-FIDELITY1 calibration

The re-audit found that introducing a 3-epoch screen creates a new scientific approximation that must be calibrated before production performance promotion. SIZE-FIDELITY1 therefore exhaustively continues all hard-coverage-qualified rungs to 30 epochs for multiple frozen screening seeds and checks whether the epoch-3 top four and epoch-10 top two retain the eventual 30-epoch winner and finalists. The 256-frame coarse monitor is compared against the full development role, and the coarse practical-equivalence width is calibrated from real trajectory/seed variability. Both current values are provisional until this evidence exists.

The continuation audit also extends beyond global RNG state: PERF-P2R must authenticate DataLoader/sampler/worker ordering state or prove deterministic epoch-boundary reconstruction. This closes a restart-equivalence gap that can otherwise survive checkpoint/optimizer/RNG digest checks.

Revised optimization roadmap

PERF-P2's lazy four-smallest truncation is superseded scientifically. The next scientific qualification gate is SIZE-FIDELITY1. PERF-P2R follows only after the epoch-3/10 fidelity and coarse-monitor calibration pass; PERF-P3 remains after PERF-P2R.

PERF-P2R must optimize the corrected semantics rather than recover the old truncation indirectly. Its scope is:

1. exact full-ladder FPS/coverage state reuse;
2. authenticated nested-corpus prefix/index views instead of duplicate bytes;
3. frame-level graph/neighbor/preprocessing cache reuse across candidate sizes;
4. target-only epoch-3 endpoint evaluation with no replay/rescue/bootstrap/ physical work;
5. exact in-place 3->10->30 pause/resume without repaying prefix epochs;
6. stage-aware single-/multi-GPU resource scheduling;
7. checkpoint retention/garbage collection after immutable elimination evidence; and
8. whole-funnel wall/VRAM/RSS/I/O/utilization qualification on an authorizing MACE/GPU runtime.

For qualifiers A, epoch-3 survivors S_4, and epoch-10 finalists S_2, the useful target exposure proxy is

$$W = 3\sum_{i\in A} K_i + 7\sum_{i\in S_4} K_i + 20\sum_{i\in S_2} K_i.$$

For all seven default sizes, $\sum K_i = 16256$. The proxy reduction relative to training all seven sizes to 30 epochs ranges from 17.56% when the largest sizes survive to 85.67% when the smallest survive. This is not a wall-time prediction.

Qualification boundary

The correction is fully testable for authority construction, serialization, role construction, continuation identity, dispatch state, and CPU-side ladder behavior in the current environment. No authorizing MACE/GPU runtime is available here, so revision 48 does **not** claim empirical epoch-3/10 survivor fidelity or measured 3/10/30 GPU training speed. Survivor-fidelity certification belongs to SIZE-FIDELITY1; performance authority belongs to PERF-P2R.

References

[1] Kevin Jamieson and Ameet Talwalkar. "Non-stochastic Best Arm Identification and Hyperparameter Optimization." *Proceedings of AISTATS*, PMLR 51:240-248, 2016. https://proceedings.mlr.press/v51/jamieson16.html

[2] Lisha Li, Kevin Jamieson, Giulia DeSalvo, Afshin Rostamizadeh, and Ameet Talwalkar. "Hyperband: A Novel Bandit-Based Approach to Hyperparameter Optimization." *Journal of Machine Learning Research* 18(185):1-52, 2018. https://www.jmlr.org/papers/v18/16-558.html